

DATA STRUCTURES (Data Structures) –

DATA STRUCTURES, ALGORITHMS, AND THEIR APPLICATIONS

1. WHAT ARE DATA STRUCTURES?

- DATA STRUCTURES ARE THE WAY WE ORGANIZE AND STORE DATA.
- DATA STRUCTURES ARE USED TO STORE AND MANIPULATE DATA.
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- Algorithm is a set of instructions to solve a problem.

2. TYPES

- Linear – Array, Linked List, Stack, Queue
- Non-Linear – Tree, Graph

3. Array

- A collection of elements stored in contiguous memory locations.
- Index starts from 0.
- Fixed size – cannot change.

4. Linked List

- Nodes are stored in non-contiguous memory locations.
- Dynamic size.
- Sequential access.

5. Stack

- LIFO – Last In First Out
- Operations – Push, Pop, Peek
- Applications – Expression evaluation

6. Queue

- FIFO – First In First Out
- Enqueue, Dequeue
- Types – Circular Queue, Priority Queue

7. Tree

- Hierarchical structure.
- Root, Parent, Child, Leaf nodes.
- Types – Binary Tree, BST, AVL

8. Graph

- Vertices ■■■■■■■■ Edges ■■■■■ ■■■■■■■■.
- Directed / Undirected.
- Traversal – BFS, DFS

9. ■■■■■■■■ ■■■■■■■■

Structure	Access	Size
Array	Fast (Index)	Fixed
Linked List	Slow	Dynamic
Stack	Top ■■■■■■■■	Dynamic
Queue	Ends ■■■■■■■■	Dynamic